

v1.0Seven-Dimensional Universe Research Request

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: G₂-Ricci Flow and 7D Manifold Architecture **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** v1.0Seven-Dimensional Universe Research Request V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^{^7}+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Swampland de Sitter	KKLT described as stable	Ooguri 2024: de Sitter Swampland challenges KKLT. 7D vacuum stability potentially inconsistent	HIGH	Theoretical foundation of V1.0 made uncertain by Swampland criterion
G ₂ topology classification	Single class assumed	Foscolo-Haskins 2024: 5 distinct ALC G ₂ manifold classes with different 4D SUSY breaking patterns	MEDIUM	Multiple topology classes available — changes effective field theory outcomes
Black hole remnant mass floor	Pre-Run 3 limits	ATLAS Run 3 2024: M _{Pl(5D)} > 11.1 TeV. BHR mass > 11.1 TeV minimum	HIGH	Core BHR dark matter parameter updated by experiment

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Torsion implementation	Analytical description	V2.0: antisymmetric Christoffel $T_{\{\mu\nu\rho\}}$ implemented as axial vector A_μ with parity-violating interaction	MEDIUM	Torsion field goes from conceptual to implemented in physics engine

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** SwampLand, Black
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Seven-Dimensional Universe Research Request V1.0 archived | V2.0 is the active specification as of June 2026